SYSTEM FLOW

How the TLC reading system works

One pipeline. It runs once for a single plate, and once per timepoint for a reaction.

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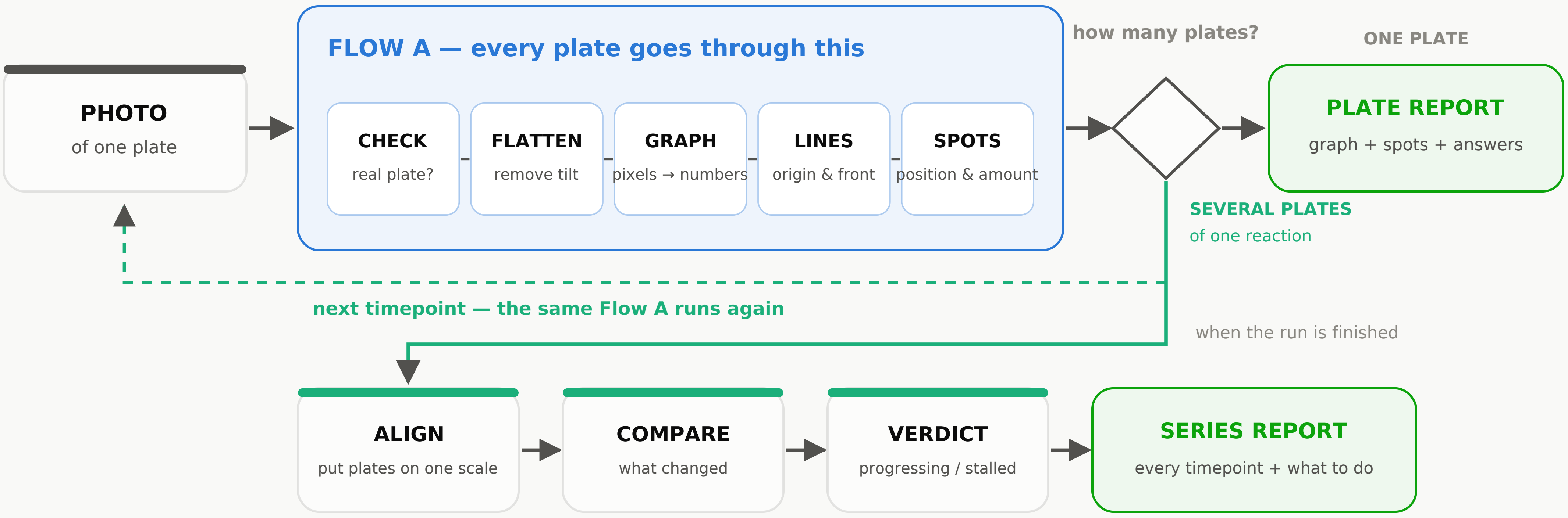
THE FLOW



The pipeline is identical either way. A series simply runs it once per plate and then compares the results.

THE ONE IDEA BEHIND BOTH PATHS

A plate photo is treated as a measurement, not a picture. Every pixel becomes a number saying how much material sits at that point, measured against the plate's own lighting. Once the plate is numbers, everything after it — where the spots are, how much of each, what changed since the last timepoint — is arithmetic that gives the same answer every time it runs.

WHAT A CHEMIST GETS BACK

From one plate	the plate as a graph, every spot with its position and share, and answers: is the starting material gone, has product formed, where are the impurities
From a series	the same for each timepoint, plus how conversion moved between them, and a verdict: progressing, stalled, complete or degrading

FLOW A · ONE PLATE

What happens to a single photo

Each step in plain language, with the reason it exists

1 · Read the photo

The picture is read exactly as the camera saved it. No brightening or sharpening is applied, because those change the very brightness values we are about to measure.

2 · Is cropping needed?

If the photo is already just the plate, nothing is cropped. Only when the plate sits inside a wider bench photo does the system cut it out.

3 · Find the plate

Under UV the plate is the brightest green object in the frame. The system finds that region and reads its four corners.

4 · Check nothing was cut off

It looks at a thin strip just outside each edge. If that strip still looks like plate, the boundary is inside the plate and is widened. This is the check that stops the system quietly deleting part of the chemistry.

5 · Straighten

The four corners are pulled into a true rectangle, so a photo taken at an angle becomes square-on and a height on the plate means the same thing everywhere.

6 · Learn the plate's own lighting

The UV lamp is never even. The system works out what this plate would look like with nothing on it — bright in some places, dimmer in others.

7 · Turn every pixel into a number

Each pixel becomes how dark it is compared with that lighting. Because it is a comparison, a brighter or dimmer photo of the same plate gives the same answer. This grid of numbers is the graph.

8 · Mark what will not be measured

The dark frame left by straightening, the heading, and the lane labels are greyed out — visible on the graph, but excluded from measurement, so a mistake here can be seen rather than hidden.

9 · Find lanes, origin and front

Lanes are placed using the lane count from the labels. The origin comes from the spotting dots. The front comes from a drawn pencil line if there is one, otherwise a working line just above the highest spot.

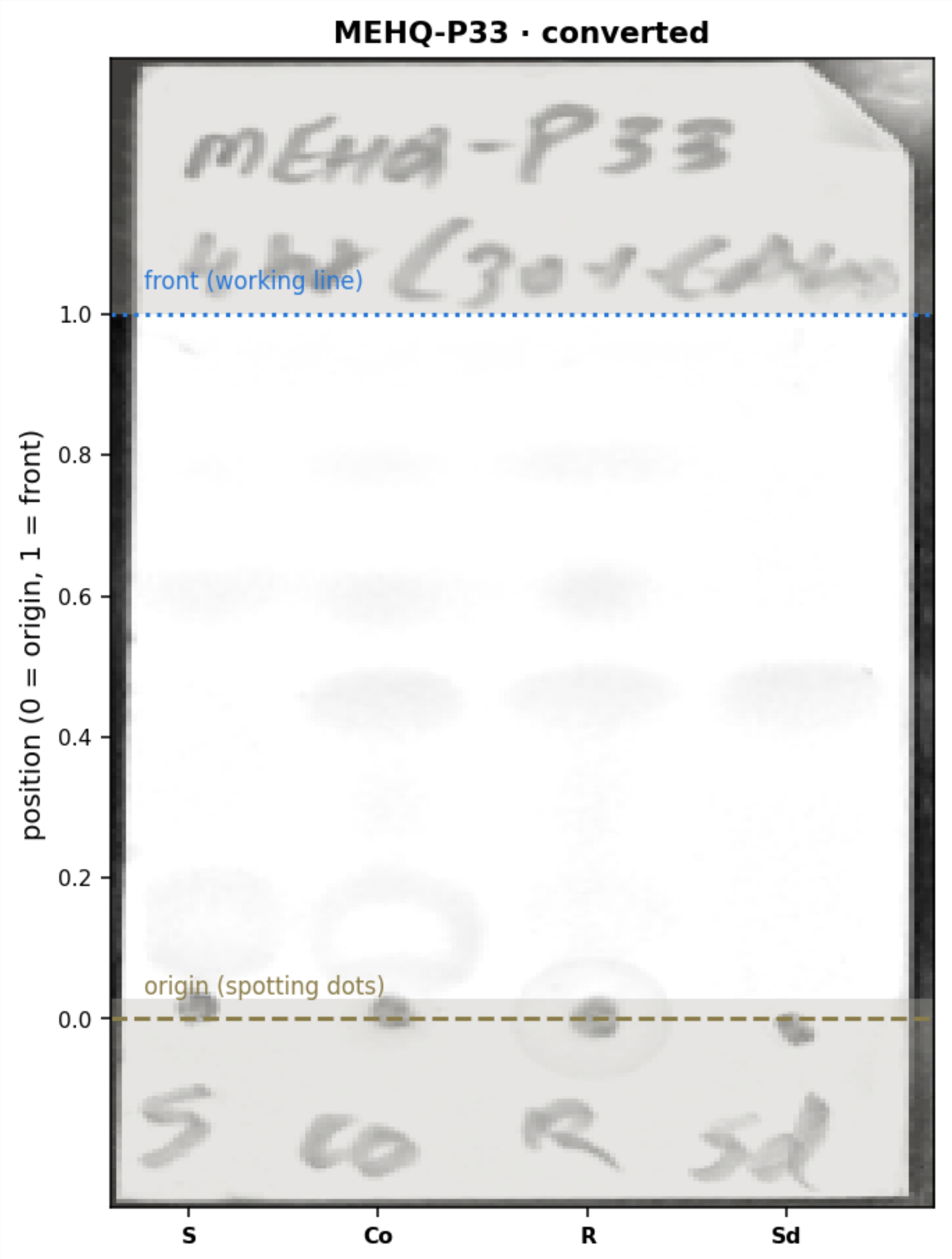
10 · Measure the spots

Each lane becomes a curve; every bump is fitted to give its position, width and how much material it holds, with an uncertainty on each.

FLOW A · OUTPUT

What comes out for one plate

A real plate from our own folder, converted by the system



The graph

Every pixel of the plate, converted. Smears and faint marks keep their exact shape; nothing is cropped away.

The two lines

Dashed = the origin, where samples were spotted. Blue = the front. Solid if a pencil line was found, dotted if the system placed a working line above the highest spot.

The spot table

For each lane, every spot with its position, how much of the lane's signal it holds, its width, and a confidence.

The answers

Is the starting material still there and how much. Has product formed. Which impurities came with the starting material rather than from the reaction. What to do next.

The warnings

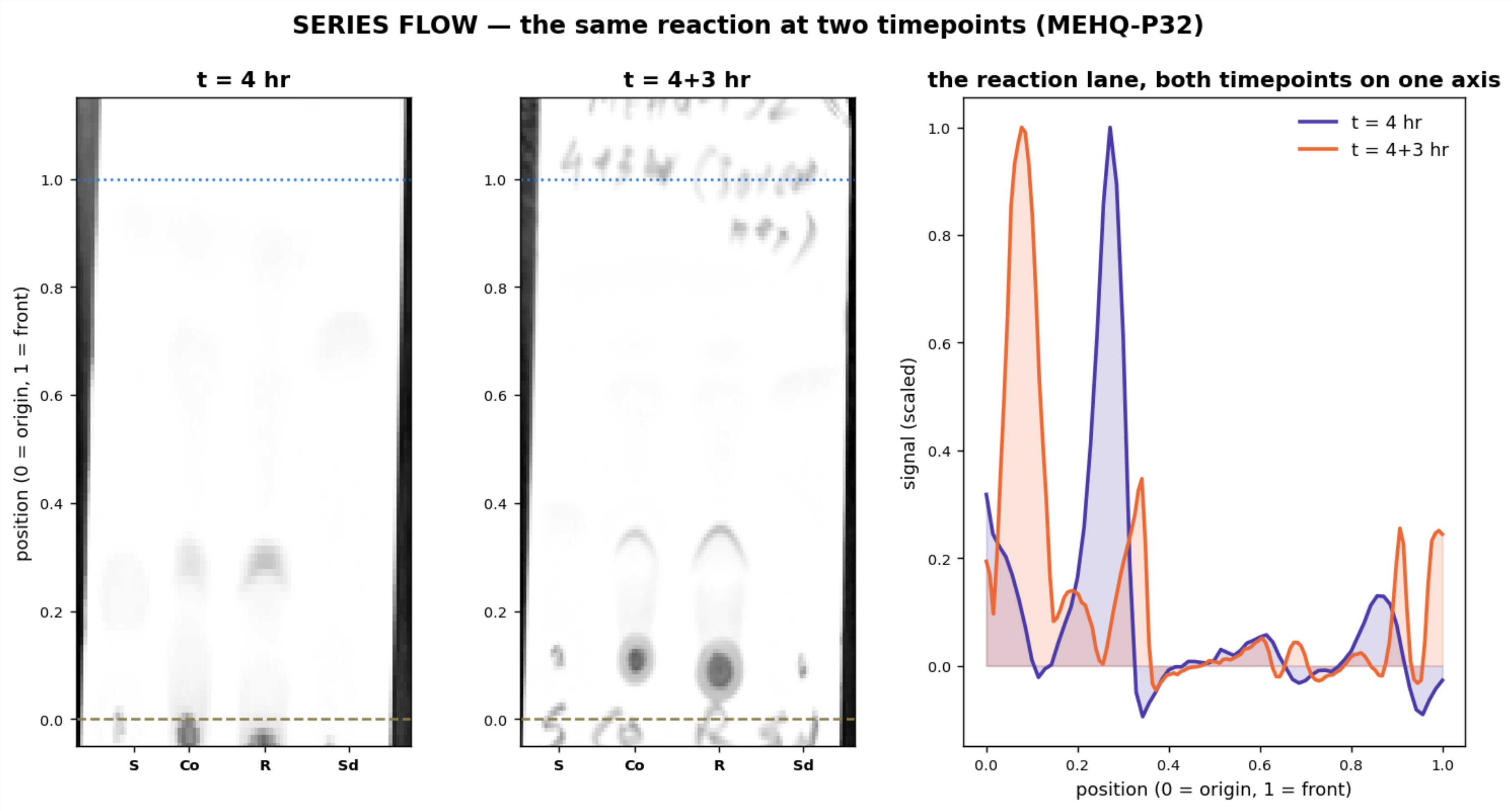
Anything that limits the result is stated on the face of it: no front line drawn, low resolution, overexposed photo, streaky lane.

REPEATABILITY

The same photo always produces the same numbers — verified by running plates twice and comparing the output byte for byte. Two chemists reading the same plate can disagree; the system cannot disagree with itself.

How several plates of one reaction are handled

Shown on two real timepoints of the same reaction from our folder



Group the plates

The chemist uploads each timepoint with its time (0 min, 30 min, 60 min…). Plates of one reaction are held together as a series.

Analyse each on its own first

Every plate runs through Flow A independently. A bad photo at one timepoint does not spoil the others.

Put them on one scale

Plates are never identical — different run distance, different exposure. Positions are lined up using the reference lanes, which contain the same material on every plate. This is what makes 'the same spot' comparable across days.

Compare timepoint to timepoint

Starting material shrinking, product growing, new spots appearing, impurities changing — measured as differences, not judged by eye.

Give a verdict

Progressing, stalled, complete, or degrading — with the numbers behind it and a confidence.

FLOW B · OUTPUT

What the chemist gets for the whole reaction

One document covering every timepoint, plus what to do next

THE FOUR VERDICTS

PROGRESSING	Starting material is falling and product is rising, by more than the measurement uncertainty. Keep going.
STALLED	Two or more intervals with no real change. Time alone will not finish it — change temperature, reagent or catalyst.
COMPLETE	Starting material is at or below the detection limit and product is steady. Stop and work up.
DEGRADING	Product is shrinking, or new spots are growing while product falls. Stop sooner rather than later.

AND ALONGSIDE THE VERDICT

Timepoint strip	Every plate as a graph, side by side on one scale, so the change is visible at a glance.
Conversion table	Apparent conversion at each timepoint with its uncertainty, and the change between consecutive ones.
What is new	Any spot that appeared at a later timepoint and was not there before, flagged for watching.
What is inherited	Impurities that were present from the very first plate — they came with the starting material, not from the reaction.
A rate estimate	If the trend is steady, an estimate of when it would finish — always labelled as an estimate, never as a promise.

EDGE CASES

When things are not ideal

What the system does, and what the chemist sees — it never guesses quietly

The photo is not a TLC plate	Rejected at the door, with the reason and the measured value.
The plate is blank	Reported as 'no spots found' rather than an empty table pretending to be a result.
The photo is overexposed	Detected. Quantities are switched off, because information destroyed by the camera cannot be recovered.
Part of the plate is outside the photo	Flagged. Positions along the cut direction are marked unreliable; Rf is switched off if the cut is at top or bottom.
No front line was drawn	A working line is placed above the highest spot, the scale is labelled 'apparent', and the chemist is told that marking the front unlocks true Rf.
No origin dots visible	Falls back to the lane-label row, with reduced confidence stated.
A lane is one long streak	Percentages for that lane are withheld, and the likely cause is reported: overloading, an acid-base effect, or decomposition on the plate.
Two spots run together	Reported as one spot with a note that its width suggests two — with the suggestion to try a weaker solvent.
Lane names are unfamiliar	The chemist is asked once, in a dropdown. The answer is remembered for the lab and never asked again.
The photo is small or blurry	Analysed anyway, with every confidence capped and the resolution limit stated.

EDGE CASES THAT ONLY EXIST IN A SERIES

A timepoint is missing or unusable	The series continues with the rest; the gap is shown rather than interpolated.
Plates were run in different solvents	Positions are not comparable. The system says so instead of drawing a trend.
Lane layout changed between plates	Each plate keeps its own lanes; only lanes with the same role are compared.
Times were not recorded	Plates are kept in upload order and the trend is shown without a time axis.
Conversion goes down then up	No smoothing is applied. The wobble is shown with its uncertainty; if it is inside the uncertainty it is reported as no real change.
Only one usable plate	It falls back to a single-plate report with a note that no trend can be given.

LIMITS AND TRUST

What it cannot do, and how you can check it

Stated upfront, because a tool that hides its limits is worse than no tool

Percentages are of signal, not moles	Different compounds glow differently under UV. A 57% product reading means 57% of the signal. Calibrated reference spots would make it exact.
True Rf needs the lines drawn	If the origin and front were not marked in pencil, positions are relative and labelled 'apparent'. One bench habit removes this limit entirely.
Very close spots cannot be split	On a small photo, two spots closer than about five percent of the plate height cannot be separated by any method.
Overexposure cannot be undone	If the camera clipped the brightness, the information is gone before the system ever sees it.
The lane count comes from you	On a low-resolution photo the system cannot reliably count lanes on its own. It asks once and remembers.

HOW A CHEMIST CAN CHECK THE SYSTEM

Every claim points at the plate	Each finding is marked on the graph, so it can be confirmed by eye in seconds.
Every number carries its reasons	Confidence is built from named factors — photo quality, whether the front was drawn, how strong the spot is.
Every run can be replayed	Each stage stores what it received, what it decided and why, so a wrong answer can be traced to the exact step.
Wrong results become tests	Anything a chemist flags is kept as a permanent test case that must keep passing after any change.